[442] NOTE SUR LA SURFACE DU QUATRIÈME ORDRE DOUÉE DE SEIZE POINTS SINGULIERS ET DE SEIZE PLANS SINGULIERS.

[From the Journal für die reine und angewandte Mathematik (Crelle), tom. LXXIII. (1871), pp. 292–293.]

L'équation de M. Kummer se transforme sans difficulté en celle-ci

$$\sqrt{\alpha x}\left(\gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}\right)+\sqrt{\beta y}\left(\alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}\right)+\sqrt{\gamma z}\left(\beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}\right)=0,$$

où

$$\alpha+\beta+\gamma=0,\quad \alpha'+\beta'+\gamma'=0,\quad \alpha''+\beta''+\gamma''=0.$$

Or cette équation rendue rationnelle prend, après toutes les réductions nécessaires, la forme suivante:

$$\begin{aligned}&w^2\big(x^2+y^2+z^2-2yz-2zx-2xy\big)\\&\quad+2w\big(\alpha\alpha'\alpha''(y^2z-yz^2)+\beta\beta'\beta''(z^2x-zx^2)+\gamma\gamma'\gamma''(x^2y-xy^2)+\theta xyz\big)\\&\quad+\big(\alpha\alpha'\alpha''yz+\beta\beta'\beta''zx+\gamma\gamma'\gamma''xy\big)^2=0,\end{aligned}$$

où, pour abréger, l'on a écrit

$$\begin{aligned}\theta&=(\beta-\gamma)\alpha'\alpha''+(\gamma-\alpha)\beta'\beta''+(\alpha-\beta)\gamma'\gamma''\\&=(\beta'-\gamma')\alpha''\alpha+(\gamma'-\alpha')\beta''\beta+(\alpha'-\beta')\gamma''\gamma\\&=(\beta''-\gamma'')\alpha\alpha'+(\gamma''-\alpha'')\beta\beta'+(\alpha''-\beta'')\gamma\gamma'\\&=-\tfrac{1}{2}\big\{(\beta-\gamma)(\beta'-\gamma')(\beta''-\gamma'')+(\gamma-\alpha)(\gamma'-\alpha')(\gamma''-\alpha'')+(\alpha-\beta)(\alpha'-\beta')(\alpha''-\beta'')\big\}.\end{aligned}$$

L'identité de ces différentes valeurs de θ étant facile à vérifier.

En représentant par $Aw^2+2Bw+C=0$ la forme rationnelle de l'équation de la surface, on trouve pour le discriminant $AC-B^2$ de cette équation du second degré en w la valeur

$$AC-B^2=4\alpha\alpha''\beta\beta'\beta''\gamma\gamma'\gamma''-2\theta\,xyz\left(\tfrac{x}{\alpha}+\tfrac{y}{\beta}+\tfrac{z}{\gamma}\right)\left(\tfrac{x}{\alpha'}+\tfrac{y}{\beta'}+\tfrac{z}{\gamma'}\right)\left(\tfrac{x}{\alpha''}+\tfrac{y}{\beta''}+\tfrac{z}{\gamma''}\right).$$

L'équation de la surface rendue rationnelle est symétrique par rapport aux trois systèmes de quantités (α,β,γ), (α',β',γ'), $(\alpha'',\beta'',\gamma'')$; la forme irrationnelle de la même équation peut donc être présentée de trois manières différentes, savoir:

$$\begin{aligned}&\sqrt{\alpha x}\left(\gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}\right)+\sqrt{\beta y}\left(\alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}\right)+\sqrt{\gamma z}\left(\beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}\right)=0,\\&\sqrt{\alpha' x}\left(\gamma''\gamma y-\beta''\beta z-\tfrac{w}{\alpha'}\right)+\sqrt{\beta' y}\left(\alpha''\alpha z-\gamma''\gamma x-\tfrac{w}{\beta'}\right)+\sqrt{\gamma' z}\left(\beta''\beta x-\alpha''\alpha y-\tfrac{w}{\gamma'}\right)=0,\\&\sqrt{\alpha'' x}\left(\gamma\gamma' y-\beta\beta' z-\tfrac{w}{\alpha''}\right)+\sqrt{\beta'' y}\left(\alpha\alpha' z-\gamma\gamma' x-\tfrac{w}{\beta''}\right)+\sqrt{\gamma'' z}\left(\beta\beta' x-\alpha\alpha' y-\tfrac{w}{\gamma''}\right)=0;\end{aligned}$$

et l'on voit de plus que les équations des seize plans singuliers sont

$$\begin{gathered}x=0,\quad y=0,\quad z=0,\quad w=0,\\ \frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=0,\quad \frac{x}{\alpha'}+\frac{y}{\beta'}+\frac{z}{\gamma'}=0,\quad \frac{x}{\alpha''}+\frac{y}{\beta''}+\frac{z}{\gamma''}=0,\\ \gamma'\gamma''y-\beta'\beta''z-\tfrac{w}{\alpha}=0,\quad \alpha'\alpha''z-\gamma'\gamma''x-\tfrac{w}{\beta}=0,\quad \beta'\beta''x-\alpha'\alpha''y-\tfrac{w}{\gamma}=0,\\ \gamma''\gamma y-\beta''\beta z-\tfrac{w}{\alpha'}=0,\quad \alpha''\alpha z-\gamma''\gamma x-\tfrac{w}{\beta'}=0,\quad \beta''\beta x-\alpha''\alpha y-\tfrac{w}{\gamma'}=0,\\ \gamma\gamma'y-\beta\beta'z-\tfrac{w}{\alpha''}=0,\quad \alpha\alpha'z-\gamma\gamma'x-\tfrac{w}{\beta''}=0,\quad \beta\beta'x-\alpha\alpha'y-\tfrac{w}{\gamma''}=0,\end{gathered}$$

les quantités α,β,γ, etc. étant liées entre elles par les trois équations

$$\alpha+\beta+\gamma=0,\quad \alpha'+\beta'+\gamma'=0,\quad \alpha''+\beta''+\gamma''=0.$$

Voilà ce me semble la forme la plus simple pour l'équation de cette surface.

Cambridge, le 23 février 1871.

[443] Note on the Solution of the Quartic Equation $\alpha U + 6\beta H = 0$.

[From the Mathematische Annalen, vol. I. (1869), pp. 54, 55.]

If U denote the quartic function $(a, b, c, d, e \mid x, y)^4$, H its Hessian

$$H = (ac - b^2,\ 2(ad - bc),\ ae + 2bd - 3c^2,\ 2(be - cd),\ ce - d^2 \mid x, y)^4,$$

α and β constants, then we may find the linear factors of the function $\alpha U + 6\beta H$ (or what is the same thing solve the equation $\alpha U + 6\beta H = 0$) by a formula almost identical with that given by me (Fifth Memoir on Quantics, *Phil. Trans.* vol. CXLVIII. (1858), see p. 446, [156]) in regard to the original quartic function U.

In fact (reproducing the investigation) if I, J are the two invariants, $M = \dfrac{I^3}{4J^2}$, Φ the cubicovariant

$$\Phi = (-a^2d + 3abc - 2b^3,\ \ldots \mid x, y)^6,$$

then the identical equation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, may be written $(1,\ 0,\ -M,\ M \mid IH, JU)^3 = -\frac{1}{4}I^3\Phi^2$, whence if $\omega_1, \omega_2, \omega_3$ are the roots of the equation $(1,\ 0,\ -M,\ M \mid \omega, 1)^3 = 0$, or what is the same thing $\omega^3 - M(6\omega - 1) = 0$, then the functions

$$IH - \omega_1 JU, \quad IH - \omega_2 JU, \quad IH - \omega_3 JU$$

are each of them a square: writing

$$\begin{aligned}(\omega_2 - \omega_3)(IH - \omega_1 JU) &= X^2,\\ (\omega_3 - \omega_1)(IH - \omega_2 JU) &= Y^2,\\ (\omega_1 - \omega_2)(IH - \omega_3 JU) &= Z^2,\end{aligned}$$

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY$, $X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \tfrac{1}{2}(\alpha + i\beta)(X - iY) + \tfrac{1}{2}(\alpha - i\beta)(X + iY) - \tfrac{1}{2}i\gamma\sqrt{X^2 + Y^2},$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3}\,\sqrt{\alpha I + 6\beta\omega_1 J},\ \ldots\ \sqrt{\omega_1 - \omega_2}\,\sqrt{\alpha I + 6\beta\omega_3 J},$$

for α, β, γ, we have

$$(\omega_2 - \omega_3)\sqrt{\alpha I + 6\beta\omega_1 J}\,\sqrt{IH + \omega_1 JU} + \ldots + (\omega_1 - \omega_2)\sqrt{\alpha I + 6\beta\omega_3 J}\,\sqrt{IH + \omega_3 JU}$$

a perfect square, and since the product of the four different values is a multiple of $(\alpha U + 6\beta H)^2$ (this is most readily seen by observing that for $\alpha U + 6\beta H = 0$, the irrational expression omitting a factor is $(\omega_2 - \omega_3)(\alpha I + 6\beta\omega_1 J) + \ldots + (\omega_1 - \omega_2)(\alpha I + 6\beta\omega_3 J)$, which vanishes identically) it follows that the expression in question is the square of a linear factor of $\alpha U + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $\alpha U + 6\beta H = 0$ are the three roots

$$\sqrt{\alpha I + 6\beta\omega_1 J}; \quad \sqrt{\alpha I + 6\beta\omega_2 J}; \quad \sqrt{\alpha I + 6\beta\omega_3 J}.$$

Cambridge, September 2, 1868.

[444] On the Centro-surface of an Ellipsoid.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2+h}+\frac{Y^2}{b^2+h}+\frac{Z^2}{c^2+h}=1,$$

$$\frac{X^2}{a^2+k}+\frac{Y^2}{b^2+k}+\frac{Z^2}{c^2+k}=1;$$

and hence

$$\begin{aligned}-\beta\gamma X^2 &= a^2(a^2+h)(a^2+k),\\ -\gamma\alpha Y^2 &= b^2(b^2+h)(b^2+k),\\ -\alpha\beta Z^2 &= c^2(c^2+h)(c^2+k),\end{aligned}$$

if for shortness

$$\alpha=b^2-c^2,\quad \beta=c^2-a^2,\quad \gamma=a^2-b^2\qquad (\alpha+\beta+\gamma=0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2+k}+\frac{Y^2}{b^2+k}+\frac{Z^2}{c^2+k}=1$$

are given by the formulae

$$\begin{aligned}-\beta\gamma a^2x^2 &= (a^2+h)^2(a^2+k),\\ -\gamma\alpha b^2y^2 &= (b^2+h)^2(b^2+k),\\ -\alpha\beta c^2z^2 &= (c^2+h)^2(c^2+k);\end{aligned}$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2+c^2>2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz, and a real intersection at Q in the plane of xy. The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP. The curve has a cusp at Q, and a node at P; viz., the curve extends beyond P, but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k), (h_1, k_1), giving the same values of (x, y, z); viz., there must exist the relations

$$(a^2+h)^2(a^2+k)=(a^2+h_1)^2(a^2+k_1),$$

$$(b^2+h)^2(b^2+k)=(b^2+h_1)^2(b^2+k_1),$$

$$(c^2+h)^2(c^2+k)=(c^2+h_1)^2(c^2+k_1);$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h,k), determines the nodal curve. But the better course is to eliminate k, k_1, thus obtaining a relation between h and h_1, in virtue whereof h_1 may be regarded as a known function of h; k and k_1 can then be readily expressed in terms of h, h_1; that is, we have k as a function of h, h_1, or in effect as a function of h. The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P=a^2+b^2+c^2,\quad Q=b^2c^2+c^2a^2+a^2b^2,\quad R=a^2b^2c^2,$$

the relation is

$$(QR+3Qh^2+Ph^4)+h_1(3Qh+4Ph^2+3h^3)+h_1^2(P+3h^2)=0;$$

this is a $(2,2)$ correspondence between the two parameters h, h_1; the united values $h_1=h$, are given by the equation $6(R+Qh+Ph^2+h^3)=0$, that is

$$(a^2+h)(b^2+h)(c^2+h)=0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ.]

[445] A Memoir on Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69. Read February 10, 1870.]

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a

10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w); this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with $16, 15, 14, 13, 12$, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P, Q, R, S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P, & Q, & R, & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} \quad ; \quad \begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

and with either arrangement we may form one and the same determinant, the Jacobian determinant $J(P, Q, R, S)$, or, equating it to zero, the Jacobian surface $J(P, Q, R, S) = 0$ of the four surfaces.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P, & Q, & R, & S, & T, & \dots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates;

viz., in the case of five surfaces, we have $J(P, Q, R, S, T) = 0$ a twofold relation representing a curve; and in the case of six surfaces, $J(P, Q, R, S, T, U) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \end{array}$$

and considering the several determinants which can be formed with any 3 or 2 columns of the matrix and equating these to zero, we have in like manner a more than onefold relation between the coordinates; viz., in the case of three surfaces, we have $J(P, Q, R) = 0$ a twofold relation representing a curve; and in the case of two surfaces $J(P, Q) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered (viz., this denotes the points common to the surfaces $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, that is, the points which are at once a node of the surface $P = 0$ and a node of the surface $Q = 0$).

If the notation were used, $J^4(P) = 0$ would denote the system of four equations $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, which are satisfied simultaneously by the coordinates (x, y, z, w) of any node of the surface $P = 0$. Although in what precedes I have used the sign $=$, it should be understood that, while $J(P, Q, R, S) = 0$ denotes a single equation, $J(P, Q, R, S, T) = 0$ or $J(P, Q) = 0$ are each of them a twofold relation, and $J(P, Q, R, S, T, U) = 0$ or $J(P, Q, R) = 0$ each of them a threefold relation. We should have a fourfold relation $J^4(P) = 0$.

4. It is not asserted that $\ldots J(P, Q, R) = 0$, $J(P, Q, R, S) = 0$, $J(P, Q, R, S, T) = 0$ form a continuous series of analogous relations; and there might even be a hiatus in using, in regard to four or more surfaces, J, and in regard to four or fewer surfaces an inverted J (viz., in regard to four surfaces, either symbol used indifferently); but there is no ambiguity in, and I have preferred to adopt, the use of the single symbol J.

5. Suppose that the orders of the surfaces $P = 0$, $Q = 0$, $\ldots$ are $a + 1$, $b + 1$, $\ldots$ (so that the orders of the differential coefficients of P, Q, $\ldots$ are a, b, $\ldots$), then we have for the orders of the several loci,

$$\begin{array}{rll} J(P,Q) = 0, & \text{point-system, order} & a^3 + a^2 b + ab^2 + b^3; \\ J(P,Q,R) = 0, & \text{curve,} & a^2 + b^2 + c^2 + bc + ca + ab; \\ J(P,Q,R,S) = 0, & \text{surface,} & a + b + c + d; \\ J(P,Q,R,S,T) = 0, & \text{curve,} & ab + ac + \ldots + de; \\ J(P,Q,R,S,T,U) = 0, & \text{point-system,} & abc + abd + \ldots + def; \end{array}$$

see, as to this, Salmon's *Solid Geometry*, Ed. 2, (1865), Appendix IV., "On the Order of Systems of Equations" [not reproduced in the later editions]. In particular, if $a = b = c \ldots = 1$, then the orders are $4, 6, 4, 10, 20$.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the orders of the terms in the diagonal be a, b, c, &c., then the number of nodes is $= \frac{1}{6}(\Sigma a \cdot \Sigma ab - 2\Sigma abc)$, or, as this may also be written, $\frac{1}{2}(\Sigma a^2 b + 2\Sigma abc)$. In particular, the formula applies to the case of the surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0,$$

(a, b, c, d) being here the orders of A, B, C, D respectively, and the orders of F, G, &c., being $\frac{1}{2}(b+c)$, $\frac{1}{2}(a+c)$, &c. If the terms are all of them linear functions of the coordinates, or $a = b = c = d = 1$, then the number of nodes is $= 10$.

7. That the surface has nodes is, in fact, clear from the consideration that any point for which the minors of the determinant all vanish will be a node; and that (for the symmetrical determinant), by making the minors all of them vanish, we establish only a threefold relation between the coordinates. The expression for the number of the nodes is, I think, obtained most readily as follows.

The nodes will be points of intersection of the curve and surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0, \qquad \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} = 0,$$

the points

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0;$$

and not only so, but they touch at the points in question; so that, multiplying together the orders of the curve and surface, and subtracting twice the order of the point-system, we obtain the expression for the number of nodes. In the particular case where the functions are all linear, we have a sextic curve and cubic surface intersecting in 18 points; but the curve and surface touch in 4 points, and the number of nodes is $(18 - 2 \cdot 4) = 10$. And in the same way the formula may be established for the general case.

8. The subsidiary theorem of the contact of the curve and surface requires, however, to be proved. Seeking for the equation of the tangent plane of the surface at any one of the points in question, we have first

$$\begin{vmatrix} \delta B, & \delta F, & \delta M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ \delta F, & \delta C, & \delta N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ \delta M, & \delta N, & \delta D \end{vmatrix} = 0,$$

where, in virtue of the equations

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

the last term vanishes. Expanding the other two terms, the equation becomes

$$D(C\delta B + B\delta C - 2F\delta F) - (N^2\delta B - 2MN\delta F + M^2\delta C) + \delta M(FN - CM) + \delta N(BN - MF) = 0;$$

in virtue of the same equations, the coefficients of δM and δN each of them vanish, and we have also

$$N^2\delta B + M^2\delta C - 2MN\delta F = \tfrac{M^2}{D}(C\delta B + B\delta C - 2F\delta F),$$

so that the equation becomes finally $C\delta B + B\delta C - 2F\delta F = 0$. Investigating by a like process the equation of the tangent of the curve

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

we find between the differentials δA, δB, &c., a twofold linear relation, expressible by means of the foregoing equation $C\delta B + B\delta C - 2F\delta F = 0$, and one other equation; that is, at each of the points in question the tangent of the curve lies in the tangent plane of the surface, or, what is the same thing, the curve and surface touch at these points.

Surfaces represented by an equation $F(P,Q) = 0$, &c.

9. In the remarks which follow as to the surfaces $F(P,Q) = 0$, $F(P,Q,R) = 0$, &c., the function F is a rational and integral function of (P,Q), (P,Q,R), &c., not in general homogeneous in regard to P, Q, R, ... but of such degrees in regard to these functions respectively as to be homogeneous in regard to the coordinates (x, y, z, w).

The surface $F(P,Q) = 0$ has in general a nodal curve $\partial_P F = 0$, $\partial_Q F = 0$; if it has besides any nodes, these are points of the point-system $J(P,Q) = 0$.

The surface $F(P,Q,R) = 0$ has in general nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$; if it has besides any nodes, these are points on the curve $J(P,Q,R) = 0$.

The surface $F(P,Q,R,S) = 0$ has not in general, but it may have, nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$, $\partial_S F = 0$; if it has any other nodes, these are points on the surface $J(P,Q,R,S) = 0$.

Nodes of a Quartic Surface; Circumscribed Cone having its vertex at a Node.

10. A quartic surface may be without nodes; or it may have any number of nodes up to 16. Consider a quartic surface having a node; and take the single node, or (if more nodes than one) any one of the nodes, as the vertex of a circumscribed cone; then, considering any plane through the vertex, the section will be a quartic curve having a node at the vertex, and the generating lines in the plane will be the tangents from the node to the quartic curve; the number of them is therefore 6, and the order of the circumscribed cone is thus $= 6$. Each tangent intersects the quartic curve in the node counting as two intersections, and in the point of contact counting as two intersections; there are consequently no singular tangents; and therefore in the circumscribed cone no singular lines arising from a singular tangency of the generating line. Hence, in the case of a single node on the surface, the circumscribed cone is a cone of the order 6 without nodal or stationary lines; and the class is $= 30$; but in the case of more than one node, say k nodes, the circumscribed cone passes through the remaining $k-1$ nodes, and the generating line through each of these nodes is a nodal line of the cone; that is, the cone has $k-1$ nodal lines, and its class is $= 30 - 2k + 2$. The cone is not of necessity a proper cone; the maximum number of nodal lines is when it breaks up into 6 planes, and we have then $k - 1 = 15$; that is, the number of nodes of the surface is at most $= 16$.

11. It is easy to form a table of the different *primâ facie* possible forms of the sextic according to the number of nodes of the surface; viz., writing 6 for a proper sextic cone without nodal lines, $6_1, 6_2, \ldots 6_{10}$ for the proper sextic cone with $1, 2, \ldots$ or 10 nodal lines; and so $5, 5_1, \ldots 5_6$ for the proper quintic cones, $4, 4_1, 4_2, 4_3, 3, 3_1, 2$ for the quartic, cubic, and quadric cones, and 1 for the plane, the table is

Circumscribed Sextic Cone.

Nodes of Surface.	
1	6
2	6_1
3	6_2
4	6_3
5	6_4
6	$6_5;\ 5, 1$
7	$6_6;\ 5_1, 1$
8	$6_7;\ 5_2, 1$
9	$6_8;\ 5_3, 1;\ 4, 2$
10	$6_9;\ 5_4, 1;\ 4_1, 2;\ 4, 1, 1;\ 3, 3$
11	$6_{10};\ 5_5, 1;\ 4_2, 2;\ 4_1, 1, 1;\ 3_1, 3$
12	$5_6, 1;\ 4_3, 2;\ 4_2, 1, 1;\ 3_2, 3;\ 3, 2, 1$
13	$\ldots;\ \ldots;\ 4_3, 1, 1;\ \ldots;\ 3, 2, 1;\ 3, 1, 1, 1;\ 2, 2, 2$
14	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ 3_1, 1, 1, 1;\ 2, 2, 1, 1$
15	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ \ldots;\ 2_1, 1, 1, 1, 1$
16	$\ldots;\ \ldots;\ \ldots;\ \ldots;\ \ldots;\ 1, 1, 1, 1, 1, 1$

and moreover, in the cases where there are two or more forms of the sextic cone, then the k sextic cones may be of the different forms in various combinations. The total number of cases *primâ facie* possible is thus very great; but only a comparatively small number of them actually exist.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane, and into a (proper or improper) quintic cone intersecting the plane in 5 lines; that is, there will be in the plane 6 nodes; the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 6 nodes lie on this conic. Taking any one of these nodes as vertex, the corresponding sextic cone breaks up into the plane, and into a (proper or improper) quintic cone.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one form of sextic cone; so that each node (at least so far as appears) stands in the same relation to the surface. Considering the last mentioned two cases; $k = 16$, each of the 16 nodes gives 6 singular tangent planes, but each of these passes through 6 nodes; therefore the number of planes is $= 16$: similarly, $k = 15$, the number of singular tangent planes is $15 \times 4 \div 6, = 10$.

For $k = 14$, the cones are $3_1, 1, 1, 1$, or $2, 2, 1, 1$: it is easy to see that we have only the three cases

Cones $3_1, 1, 1, 1$: $2, 2, 1, 1$		Singular tangent planes
may be 14, 0	gives	$(14 \cdot 3 + 0 \cdot 6) \div 6, = 7$
8, 6	„	$(8 \cdot 3 + 6 \cdot 2) \div 6, = 6$
2, 12	„	$(2 \cdot 3 + 12 \cdot 2) \div 6, = 5$

and we may in the like manner limit the number of possible cases, for other values of k. But I do not at present further pursue the inquiry.

As to the Number of Constants contained in a Surface.

14. We say that a surface $P = 0$ contains or depends upon a certain number of constants; viz., this is the number of constants contained in the equation $P = 0$ of the surface, taking the coefficient of any one term to be equal to unity; thus the general quadric surface contains 9 constants; the surface can in fact be determined so as to satisfy 9 conditions; or, as we might express it, the Postulation of the surface is $= 9$. [I have elsewhere said *Postulandum* and *Capacity*: I prefer this last expression.] And if, in the general equation so containing 9 constants,

k of these are given, or, what is the same thing, if the quadric surface be made to satisfy any k conditions, then the number of constants, or postulation of the surface, is $= 9 - k$.

15. But a different form of expression is sometimes convenient; the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0$, $Q = 0$, ... they will be satisfied by the surface $\alpha P + \beta Q + \ldots = 0$, where $\alpha, \beta, \ldots$ are any constant multipliers whatever. When this is so, there will be a certain number of solutions $P = 0$, $Q = 0$, ... not connected by any such relation, or say of asyzygetic solutions, such that the general surface satisfying the conditions in question is $\alpha P + \beta Q + \ldots = 0$; and hence, taking one of these coefficients as unity, the number of constants, or postulation of the surface, is equal to the number of the remaining coefficients, or, what is the same thing, it is less by unity than the number of the asyzygetic solutions $P = 0$, $Q = 0, \ldots$. Instead of considering the number of constants, or postulation, we may consider the number of solutions (that is, asyzygetic solutions) or surfaces $P = 0$, $Q = 0$, ... which satisfy the conditions in question.

16. Thus, for the quadric not subjected to any conditions, there are 10 surfaces (for example, these may be taken to be the surfaces $x^2 = 0$, $y^2 = 0$, $z^2 = 0$, $w^2 = 0$, $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$); and the general quadric surface is by means of these expressed linearly in the form $(a, \ldots \mid x, y, z, w)^2 = 0$. So for the quadric surfaces through k given points, the number of these is $= 10 - k$; thus for the surfaces through 4 given points, say the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, the 6 given surfaces may be taken to be $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$, and every other quadric surface through the 4 points is by means of these expressed linearly in the form $(a, \ldots \mid yz, zx, xy, xw, yw, zw) = 0$; for the quadric surfaces through 8 points there are two surfaces $P = 0$, $Q = 0$; and every quadric surface through the 8 points is by means of these expressed linearly in the form $\alpha P + \beta Q = 0$; and (as the extreme case) if the quadric surface passes through 9 given points, then there is the single quadric surface $P = 0$.

17. In the questions in which such quadric surfaces present themselves, it is in general quite immaterial what particular surfaces are selected as the surfaces $P = 0$, $Q = 0, \ldots$; the selection may be made at pleasure; and, being so made, the surfaces are to be regarded as completely determinate; viz., there would be no gain of generality if these were replaced by any other surfaces $\alpha P + \beta Q + \ldots = 0$. For instance, in the theory of the quartic surfaces with 6 given points as nodes, we have through the 6 given points the 4 quartic surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$, and we consider the quartic functions $(a, \ldots \mid P, Q, R, S)^2$ and $J(P, Q, R, S)$: each of these is unaltered as to its form when P, Q, R, S are replaced each of them by any linear function of these quantities; viz., $(a, \ldots \mid P, Q, R, S)^2$ is changed into a new quadric function $(a', \ldots \mid P, Q, R, S)^2$, and $J(P, Q, R, S)$ into a mere constant multiple of its original value. We have herein a justification of the expressions in question, through 6 given points there are 4 quadric surfaces, &c.

General theory of the Quartic Surface with a given Node or Nodes.

18. A quartic surface contains 34 constants; and the number of conditions to be satisfied in order that a given point may be a node is $= 4$. Hence, if the surface has k given points as nodes, the number of constants is $= 34 - 4k$; and it would at first sight appear that k might be $= 8$, and that with the 8 given points as nodes we should have a quartic surface containing 2 constants. But this is not so in a proper sense; through the 8 given points we have 2 quadric surfaces $P = 0$, $Q = 0$; and we can by means of these form a quartic surface $(a, b, c \mid P, Q)^2 = 0$, containing 2 constants, and having in a sense the 8 points as nodes. This, however, is no proper quartic surface, but is a system of 2 quadric surfaces, each of them passing through the 8 points, and the two quadric surfaces therefore intersecting in a quadri-quadric curve through the 8 points; which curve is therefore a nodal curve on the compound surface; and it is only as points on this nodal curve, and not in a proper sense, that the 8 given points are nodes of the quartic surface. The greatest value of k is thus $k = 7$.

19. Of course, if $k = 0$, we have the general quartic surface $U = 0$, containing 34 constants. The cases $k = 1$, $k = 2$, $k = 3$ (viz., a single given node, 2 given nodes, 3 given nodes), may be at once disposed of; taking for instance the 1st node to be the point $(1, 0, 0, 0)$, the 2nd node the point $(0, 1, 0, 0)$, the 3rd node the point $(0, 0, 1, 0)$, we find at once an equation $U = 0$, with $30, 26$, or 22 constants, having the given node or nodes.

Four given Nodes.

20. The case of 4 given nodes is just as easy; but in reference to what follows, it is proper to consider it more in detail. The equation should contain 18 constants; we have through the 4 given points 6 quadric surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, $U = 0$, and we can by means of them form a quartic equation $(a, \dots \mid P, Q, R, S, T, U)^2 = 0$, having the 4 given points as nodes; this contains, however, $(21 - 1 =)20$ constants; the reduction to the right number 18 occurs by reason that the functions (P, Q, R, S, T, U), although linearly independent, are connected by two quadric equations

$$(* \mid P, Q, R, S, T, U)^2 = 0, \qquad (*' \mid P, Q, R, S, T, U)^2 = 0;$$

hence writing the equation of the quartic surface in the form

$$(a, \dots \mid P_*)^2 - \lambda\,(* \mid P_*)^2 - \mu\,(*' \mid P_*)^2 = 0,$$

the coefficients λ, μ may be so determined as to reduce to zero the coefficients of any two terms of the equation, and the number of constants really is $20 - 2 = 18$, as it should be.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, the quadric surfaces may be taken to be $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$; the equation of the quartic surface will thus be

$$(a, \dots \mid yz, zx, xy, xw, yw, zw)^2 = 0;$$

but we have between the functions xy, &c., the two identical relations

$$xy \cdot zw - xz \cdot yw = 0, \qquad xy \cdot zw - xw \cdot yz = 0;$$

and the number of constants is thus $= 18$.

Five given Nodes.

22. In the case of 5 given nodes, the number of constants should be $= 14$. We have through the 5 given points, 5 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, and we form herewith the quartic equation $(a, \dots \mid P, Q, R, S, T)^2 = 0$, containing the right number 14 of arbitrary constants. The functions P, Q, &c. are in this case not connected by any quadric relation, and the equation just written down is in fact the general equation of the quartic surface with the 5 given nodes.

23. In verification, take the first 4 nodes to be as above, and the 5th node to be the point $(1, 1, 1, 1)$; we may write

$$(P, Q, R, S, T) = \{x(y-z),\ x(y-w),\ y(x-z),\ y(x-w),\ xy-zw\};$$

and if from the 5 equations $P = x(y-z)$, &c., we eliminate (x, y, z, w), we obtain one, and only one, relation between the functions P, Q, R, S, T; this is found to be

$$PS(Q+R-T) - QR(P+S-T) = 0,$$

or, what is the same thing,

$$R(P-Q)(S-T) - P(R-S)(Q-T) = 0;$$

viz., it is a cubic relation, and there is consequently no quadric relation between the 5 functions.

Six given Nodes.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points 4 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$; but if we form herewith the quartic surface $(a, \dots \mid P, Q, R, S)^2 = 0$, this contains only 9 constants. It is to be shown that the Jacobian surface $J(P, Q, R, S) = 0$ of the 4 quadric surfaces (or say of the 6 points) is a quartic surface having the 6 given points as nodes, and not included in the foregoing form $(a, \dots \mid P, Q, R, S)^2 = 0$; this being so, we have the quartic surface

$$(a, \dots \mid P, Q, R, S)^2 + \theta\, J(P, Q, R, S) = 0,$$

having the 6 given points as nodes, and containing the complete number of constants, viz., 10.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, $(1, 1, 1, 1)$, and $(\alpha, \beta, \gamma, \delta)$. I proceed to find the Jacobian of these 6 points. For this purpose, let (a, b, c, f, g, h) be the 6 coordinates of the line through the points $(1, 1, 1, 1)$ and $(\alpha, \beta, \gamma, \delta)$, viz.,

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, \\ b &= \gamma - \alpha, & g &= \beta - \delta, \\ c &= \alpha - \beta, & h &= \gamma - \delta, \end{aligned}$$

whence $af + bg + ch = 0$, and also

$$\begin{array}{rrrr} 0\cdot & -h\cdot & -g\cdot & +a = 0, \\ -h\cdot & +0\cdot & +f\cdot & +b = 0, \\ g\cdot & -f\cdot & +0\cdot & +c = 0, \\ -a\cdot & -b\cdot & -c\cdot & +0 = 0; \end{array}$$

we have through the 6 points the plane pairs

$$\begin{array}{l} x\,(\qquad\ \dots - hy - gz + aw) = 0, \\ y\,(-hx \qquad\ + fz + bw) = 0, \\ z\,(\ \ gx - fy \qquad\ + cw) = 0, \\ w\,(-ax - by - cz \qquad\ ) = 0, \end{array}$$

where, adding the four equations, we have identically $= 0$. For this reason, we cannot take these to be the equations of the 4 quadric surfaces, but we may take the first 3 of them for the surfaces

$P=0$, $Q=0$, $R=0$; and for the 4th surface $S=0$, I take the quadric cone having its vertex at the point $(0,0,0,1)$; viz., the equation is

$$a\alpha\, yz + b\beta\, zx + c\gamma\, xy = 0;$$

that is, I write

$$(P,Q,R,S) = \{x(hy-gz+aw),\ y(-hx+fz+bw),\ z(gx-fy+cw),\ (a\alpha yz+b\beta zx+c\gamma xy)\}.$$

26. The Jacobian is then easily found to be

$$\begin{aligned}&(b\beta zx+c\gamma xy)(-agh, bhf, cfg, abc, -af^2, -gB, hC, aA, bfg, -c^2h \mid x,y,z,w)^2\\ &\quad+(c\gamma xy+a\alpha yz)(agh, -bhf, cfg, abc, fA, -bg^2, -hC, -a^2f, bB, c^2h \mid x,y,z,w)^2\\ &\quad+(a\alpha yz+b\beta zx)(agh, bhf, -cfg, abc, -fA, gB, -ch^2, a^2f, -b^2g, cC \mid x,y,z,w)^2=0;\end{aligned}$$

where for the moment A, B, C denote $bg-ch$, $ch-af$, $af-bg$ respectively. Collecting and reducing, the whole divides by $2abc$; and if finally we replace a,b,c,f,g,h by their values, the result is

$$J=\left\{\begin{aligned}&(\beta-\gamma)yz\,(\alpha w^2-\delta x^2)+(\alpha-\delta)xw\,(\beta z^2-\gamma y^2)\\ &\quad+(\gamma-\alpha)zx\,(\beta w^2-\delta y^2)+(\beta-\delta)yw\,(\gamma x^2-\alpha z^2)\\ &\quad+(\alpha-\beta)xy\,(\gamma w^2-\delta z^2)+(\gamma-\delta)zw\,(\alpha y^2-\beta x^2)\end{aligned}\right\}=0.$$

27. It may be shown *à posteriori* that J is not a quadric function of P,Q,R,S. For, attempting to express it in this form, J does not contain the terms x^2w^2, y^2w^2, z^2w^2, and it thence at once appears that the coefficients of P^2, Q^2, R^2 each of them vanish. Hence, introducing for convenience the factor 2, I assume $(0,0,0,D,F,G,H,L,M,N \mid P,Q,R,S)^2 = 2J$. Comparing the terms in $w^2(yz,zx,xy)$, we obtain

$$bcF=a\alpha,\quad caG=b\beta,\quad abH=c\gamma;$$

and comparing the coefficients of $w(y^2z, z^2x, x^2y, yz^2, zx^2, xy^2)$, we obtain

$$\begin{aligned}-Ff+a\alpha M&=\tfrac{a\alpha}{c}\beta,\quad & Ff+a\alpha N&=-\tfrac{a\alpha}{b}\gamma,\\ -Gg+b\beta N&=\tfrac{b\beta}{a}\gamma,\quad & Gg+b\beta L&=-\tfrac{b\beta}{c}\alpha,\\ -Hh+c\gamma L&=-\tfrac{c\gamma}{b}\alpha,\quad & Hh+c\gamma M&=-\tfrac{c\gamma}{a}\beta;\end{aligned}$$

substituting for F,G,H their values, we obtain from the first 3 equations $L,M,N=\frac{f}{bc}, \frac{-g}{ca}, \frac{-h}{ab}$, and from the second 3 equations, $L,M,N=\frac{f}{bc}, \frac{g}{ca}, \frac{h}{ab}$; that is, the equations are inconsistent, and the function J is not expressible in the form in question.

Jacobian Surface of Six given Points.

28. The equation $J=0$ is the locus of the vertices of the quadric cones which pass through the given 6 points; calling these $1,2,3,4,5,6$, we see at once that the surface passes through the 15 lines $12, 13, \ldots 56$, and also through the ten lines 123.456 (viz., line of intersection of the planes through $1,2,3$, and through $4,5,6$), &c. In fact, taking the vertex at any point 0 in the line $1,2$, the lines drawn to the six points are $01=02, 03, 04, 05, 06$; viz., there are only five lines, so that these lie in a quadric cone. And taking the vertex at any point in the line 123.456, the lines to the 6 points lie in these planes 123 and 456 respectively, and the quadric cone is in fact this plane-pair. Moreover, the surface containing the lines $12, 13, 14, 15, 16$, must have the point 1 for a node; and similarly, the points $2,3,4,5,6$ are each of them a node on the surface. It is to be added that the surface contains the skew cubic through the 6 points, or say the skew cubic 123456. See, as to this, *post* No. 108.

29. The surface in question (the Jacobian of the 6 points) is a particular case of the Jacobian of any 4 quadric surfaces. This more general surface will be considered in the sequel; I only remark here that it contains 10 lines, corresponding to the 10 lines 123.456, &c., but it has not any other lines, or any nodes.

Jacobian Curve of Seven given Points, or of an Octad of Points.

30. In connexion with what precedes, we may here consider a curve which presents itself in the sequel; viz., the curve which is the locus of the vertices of the quadric cones which pass through seven given points. The general case is when no one of the points is the vertex of a quadric cone through the other 6 points. We have through the 7 points the three quadric surfaces $P = 0$, $Q = 0$, $R = 0$; hence, forming the equation $\alpha P + \beta Q + \gamma R = 0$ of the general quadric surface through the 7 points, and making this a cone, we find as the locus of the vertex $J(P, Q, R) = 0$; the analytical form shows that this is a sextic curve. It appears, moreover, that the curve is symmetrically related to all the 8 points $P = 0$, $Q = 0$, $R = 0$; and instead of calling it the Jacobian of the 7 points, we may call it the Jacobian of the octad. But in further explanation, take the points to be $1, 2, 3, 4, 5, 6, 7$; the vertex will lie on each of the Jacobian surfaces 123456 and 123457; and it is at present assumed that 7 is not a point on the first surface, nor 6 a point on the second surface. The two surfaces have in common the lines $12, 13, \ldots 45$, and they consequently besides intersect in a curve of the 6th order, or sextic curve, which is the locus in question. At the point 1 there is on the first surface a tangent cone through the lines $12, 13, 14, 15, 16$, and on the second surface a tangent cone through the lines $12, 13, 14, 15, 17$; these two cones have for their complete intersection the lines $12, 13, 14, 15$, which lines belong to the complete intersection of the two surfaces, but not to the sextic curve. It thus appears, *à posteriori*, that the sextic curve does not pass through the point 1; and similarly, that it does not pass through any of the points $2, 3, 4$, or 5. As to the points 6 and 7, each of these is on only one of the quartic surfaces, and therefore the curve of intersection does not pass through either of these points.

31. Suppose, however, that one of the seven points is the vertex of a cone through the other six; it is of course the same thing whether we take this to be one of the points $1, 2, 3, 4, 5$, or one of the points 6 and 7, but the result comes out more easily in the latter case; viz., in the former case, taking 1 to be the point in question, the two tangent cones at 1 are one and the same cone, and all that appears is that there is nothing to hinder a branch or branches of the sextic curve from passing through the point 1. But in the latter case, taking 7 for the point in question, then 7 lies on the surface 123456, being a simple point on this surface, but a node on the surface 123457; and it thus appears that there are through 7 two branches of the sextic curve; so that any one of the seven points, being the vertex of a cone through the other six, is an actual double point on the sextic curve.

32. In the case where two of the points are each of them the vertex of a cone through the other six points, then the seven points lie on a skew cubic; and the sextic curve of the general case becomes this skew cubic twice repeated.

Seven given Nodes.

33. In the case of 7 given nodes, the number of constants should be $= 6$; the 7 given points determine 3 quadric surfaces $P = 0$, $Q = 0$, $R = 0$: and we have hence the quartic surface $(a, \ldots \mid P, Q, R)^2 = 0$, containing 5 constants only. That this is not the general quartic surface with the 7 given nodes, is also clear from the consideration that the surface in question has 8 nodes; viz., the 8 points of intersection of the three quadric surfaces. Suppose that a particular quartic surface, having the 7 given nodes, but not of the last mentioned form, is $\Delta = 0$; then a quartic surface having the 7 given nodes is

$$(a, \dots \mid P, Q, R)^2 + \theta\,\Delta = 0;$$

and this, as containing 6 constants, will be the general quartic surface with the 7 given nodes.

34. It follows that, if $\Delta' = 0$ be another quartic surface having the 7 given nodes, we must have identically $\Delta' - \rho\Delta = (* \mid P, Q, R)^2$, where ρ is a determinate constant and $(* \mid P, Q, R)^2$ a determinate quadric function of (P, Q, R). The formula extends to the case where $\Delta' = 0$ has the 8 nodes ($P = 0$, $Q = 0$, $R = 0$), but we have then $\rho = 0$, and the meaning is simply that the general quartic surface having the 8 nodes is $(* \mid P, Q, R)^2 = 0$.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, but not having the 8th node, is $M\Omega = 0$, where $M = 0$ in the plane through any 3 of the 7 points and $\Omega = 0$ is the cubic surface through these same 3 points, and having the remaining 4 points as nodes. The equation of the cubic surface, if the 4 points are taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, is obviously of the form

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} + \frac{d}{w} = 0, \quad (\text{that is, } ayzw + bzxw + cxyw + dxyz = 0),$$

and by making the surface pass through the 3 points we determine linearly the coefficients (a, b, c, d), that is, their ratios. The equation of the quartic surface thus is

$$(a, \dots \mid P, Q, R)^2 + \theta\,M\Omega = 0,$$

the 7 given points being here proper nodes; and the formula being precisely equivalent to the preceding one containing Δ.

36. We can with the 7 given points form 35 such combinations $M\Omega = 0$ of a plane and a cubic surface, and so present the equation of the quartic surface under 35 different forms; these are of course equivalent in virtue of the before mentioned formula for $\Delta' - \rho\Delta$; viz., we must have identically $M\Omega - \rho' M'\Omega' = (* \mid P, Q, R)^2$: a theorem of some interest, which it might be difficult to verify *à posteriori*.

Investigation of the cases of 8 Nodes.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the surface with 7 given nodes; viz.,

$$(a, \dots \mid P, Q, R)^2 + \theta\,\nabla = 0,$$

or, what is the same thing,

$$(a, \dots \mid P, Q, R)^2 + \theta\,M\Omega = 0;$$

and we inquire in what cases this surface has an 8th node. Obviously if $\theta = 0$, that is, if the surface is $(a, \dots \mid P, Q, R)^2 = 0$, the surface will have an 8th node, the remaining intersection of the quadric surfaces $P = 0$, $Q = 0$, $R = 0$ (observe that this is a point in no wise depending on the particular quadric surfaces, but uniquely determined by means of the 7 given points); and we have thus one kind, say the "octadic" surface, of the quartic surfaces with 8 nodes; viz., the nodes are the 8 points of intersection of any 3 quadric surfaces, or they are an octad of points. By what precedes, 7 of the nodes may be given points, and the remaining node is then a uniquely determinate point, the 8th point of the octad.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then be a point on the Jacobian surface $J(P, Q, R, \nabla) = 0$, which is of the order 6. It is clear *à priori* that this must be a surface depending only on the 7 points, but independent of the particular surfaces $P = 0$, $Q = 0$, $R = 0$, $\nabla = 0$; to verify this, observe that, substituting for ∇ the function $\nabla_1 = \nabla + (* \mid P, Q, R)^2$, we in fact leave the Jacobian unaltered; I call it the dianodal surface of the 7 points.

39. I say that the 8th node may be any point whatever on the dianodal surface; in fact, regarding for a moment the coordinates of the node as given, and expressing that the point is a node on the quartic surface, we have 4 equations containing

$$aP_0 + hQ_0 + gR_0, \quad hP_0 + bQ_0 + fR_0, \quad gP_0 + fQ_0 + cR_0,$$

(P_0, Q_0, R_0 the values of P, Q, R at the node,) but which, if only the point be on the dianodal surface, reduce themselves to three equations; viz., we have between the coefficients (a, b, c, f, g, h) three equations which being satisfied, the point in question will be a node. And it thus appears that, taking the 8th node to be a given point on the dianodal surface, the equation $(a, \ldots \mid P, Q, R)^2 + \theta\,\nabla = 0$ of the quartic surface will contain 3 constants. Observe that we may through the 8 nodes draw 2 quadric surfaces $P = 0$, $Q = 0$; and this being so if $\Delta = 0$ be a particular quartic surface with the 8 nodes, then the general quartic surface will be

$$(a, b, c \mid P, Q)^2 + \theta\,\Delta = 0,$$

containing the right number 3 of constants. But there is not here any simple form of the surface $\Delta = 0$, such as the form $M\Omega = 0$ for the surface through 7 given points.

40. It is clear *à priori* that the relation between the 8 nodes is a symmetrical one; so that the 8th point being situate anywhere on the dianodal surface of the remaining 7 points, each of the 7 points will be situate on the dianodal surface of the remaining 7 points. This is a remarkable property of the dianodal surface, which will have to be again considered.

41. In what precedes, we have the second kind of quartic surfaces with 8 nodes, say the "dianome"; viz., each node is a point on the dianodal surface of the remaining nodes; any 7 of the nodes may be taken to be given points, and the remaining node to be any point whatever on the dianodal surface of the 7 points.

The Dianodal Surface.

42. Consider the seven points $1, 2, 3, 4, 5, 6, 7$. As already mentioned, through three of these say $1, 2, 3$, we may draw a plane $M = 0$; and through the same three points and the remaining points $4, 5, 6, 7$ as nodes ($3 + 4 \cdot 4 = 19$ conditions), a cubic surface $\Omega = 0$; this surface passes through the six lines $45, 46, \ldots 67$. Hence we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines $12, 13, \ldots 67$, and through 35 plane cubics such as $M = 0$, $\Omega = 0$; viz., this is a cubic in the plane 123 passing through the points $1, 2, 3$, and through the intersection of the plane with each of the six lines $45, 46, \ldots 67$ (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines $12, 13, 23$. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega)\,M + J(P, Q, R, M)\,\Omega = 0$$

is satisfied by $M = 0$, $\Omega = 0$; and for the passage through the lines, taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the planes $567, 674, 745$, and 456 respectively, each of the functions

P, Q, R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0$, $w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} cy, & c'y, & c''y, & * \\ cx, & c'x, & c''x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0$, $w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes $712, 713, \ldots 756$ in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections $1, 2, 3, 4, 5, 6$, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points $7, 1, 2$ and the six intersections of the plane with the lines $34, 35, \ldots 56$. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points $1, 2, 3, 4, 5, 6$ and another point $7'$. As seen above, we have in general, through the points $1, 2, 7'$ and with the points $3, 4, 5, 6$ as nodes, a determinate cubic surface, which surface passes through the lines $34, 35, \ldots 56$. But the cubic surface becomes indeterminate if the points $1, 2, 7', 3, 4, 5, 6$ are on the same skew cubic; that is, if $7'$ is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, $7'$ as the intersection of the skew cubic by the plane 712, we have in this plane the points $7', 1, 2$, and the intersections of the plane by the lines $34, 35, \ldots 56$, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points $7, 1, 2$ and the six intersections will pass through the point $7'$; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ as the equations of the planes $456, 563, 634, 345$ respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of $7'$; the equation of the cubic surface passing through $7', 1, 2$, and having the nodes $3, 4, 5, 6$, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{1}{x_1}, & \frac{1}{y_1}, & \frac{1}{z_1}, & \frac{1}{w_1} \\ \frac{1}{x_2}, & \frac{1}{y_2}, & \frac{1}{z_2}, & \frac{1}{w_2} \end{vmatrix} = 0;$$

viz., considering (x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) as given, this is a twofold relation between the coordinates (x, y, z, w) of the point $7'$. The relation may be represented by the four equations $(yzw) = 0$, $(zwx) = 0$, $(wxy) = 0$, $(xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0$, $(zwx) = 0$ in the line $z = 0$, $w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point $7'$, and which, as is seen at once, passes through the points $1, 2, 3, 4, 5, 6$.

46. By what precedes, we have on the dianodal surface through the point 1 the lines $12, 13, 14, 15, 16, 17$, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines $12, 13, 14, 15, 16$, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines $12, 13, 14, 15, 16$.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points $1, 2, 3$, and $\Omega = 0$ the cubic surface through these points, and having the points $4, 5, 6, 7$, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1), &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1), and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x's show in like manner the orders in regard to the current coordinates (x, y, z, w); the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w), or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w), it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (xx_1x_2x_3x_4x_5x_6x_7)^6 = 0$.

48. It is possible that taking the 4 points, say $1, 2, 3, 4$, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say $5, 6, 7$, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the [paper 445 continues on book p. 151].